

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

May June 2023

20ECE42 – ANALOG AND MIXED MODE VLSI DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** and **Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Assuming transistors in saturation and $\lambda_n = \lambda_p = 0$, Draw the small signal model for the circuit shown in **Figure Q1a**. and calculate the gain

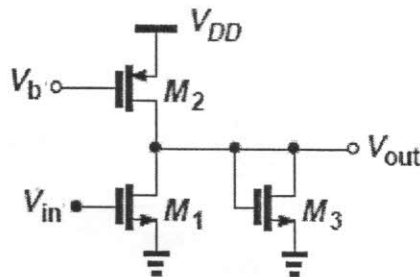


Figure Q1a.

10 L*3 1 1

- b) Draw the circuit diagram of an unbalanced single stage operational amplifier. Explain its operation.

10 L2 2 1

2. a) Assuming transistors in saturation and $\lambda_p \neq 0$ and $\lambda_n \neq 0$, Draw the small signal model for the circuit shown in **Figure Q2a**. and calculate the gain.

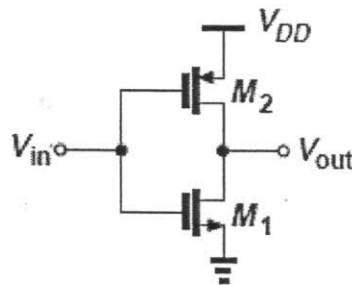


Figure Q2a.

10 L3 1 1

- b) The MOS differential pair must be designed for an equilibrium overdrive of 200 mV. If $\mu_n C_{ox} = 100 \mu A/V^2$ and $W/L = 20/0.18$, what is the required value of I_{SS} ? What would be the overdrive voltage if tail current is doubled?

4 L3 2 2

- c) Draw the circuit diagram for a simple two stage operational amplifier. What is the gain provided by each stage and the overall gain of the configuration.

6 L2 2 1

3. a) Assuming transistors in saturation and $\lambda_p \neq 0$ and $\lambda_n \neq 0$, Draw the small signal model for the circuit shown in **Figure Q3a.** and calculate the gain.

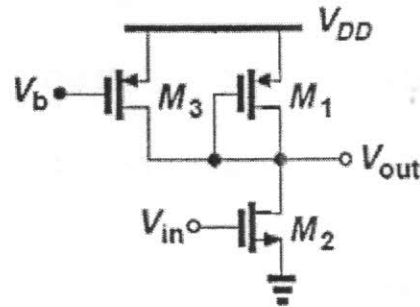


Figure Q3a

10 L3 1 1

- b) Explain the working of a current mirror and derive its equation. Also Calculate I_{copy} in the circuit shown in **Figure Q3b**

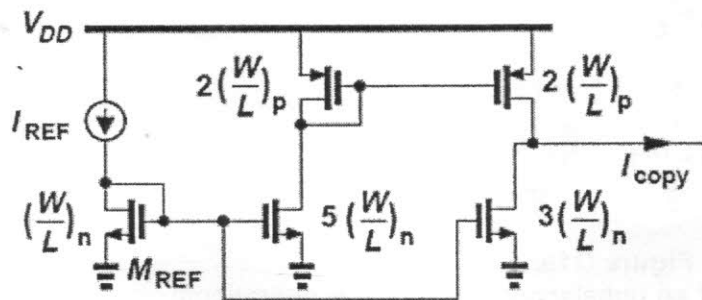


Figure Q3b

10 L3 2 2

Unit - II

4. a) With a neat diagram explain the working of 4-bit Successive Approximation ADC, clearly listing the conversion process.
- b) Determine the DNL and INL (in LSB units) for the 3-bit DAC with the given output values. Assume $V_{REF}=5V$

10 L2 4 1

Digital Code	Analog Value (V)
000	0
001	0.625
010	1.5625
011	2
100	2.5
101	3.125
110	3.4375
111	4.375

6 L3 3 2

- c) With Respect to DAC define
- Resolution.
 - Integral Non Linearity
5. a) With a neat diagram explain the working of 2-Step Flash ADC.
- b) For a 6-bit Charge Scaling Split Array DAC, Draw the equivalent circuit for $D_5D_4D_3D_2D_1D_0 = 10000$ and $D_5D_4D_3D_2D_1D_0 = 000001$ and calculate the output voltages.
- c) With Respect to DAC define
- Differential Non-Linearity
 - Least Significant Bit

4 L2 3 1

10 L2 4 1

6 L3 3 2

4 L2 3 1

- | | | | | | | |
|----|-----|--|----|----|---|---|
| 6. | a) | With a neat diagram explain the working of Dual Slope ADC. | 10 | L2 | 4 | 1 |
| | b) | For an 8-bit pipelined ADC, all the amplifiers had a gain of 2.1V/V instead of 2V/V. If $V_{in}=3V$ and $V_{ref}=5V$, what would be the resulting digital output assuming all other components are ideal. | 6 | L3 | 3 | 2 |
| | c) | With Respect to DAC define. | | | | |
| | i) | Offset Error. | 4 | L2 | 3 | 1 |
| | ii) | Dynamic Range | | | | |

Unit – III

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|----|----|--|----|----|---|---|
| 7. | a) | With Neat Sketches explain the working of Simple Delay element using Clocked CMOS. | 10 | L2 | 5 | 1 |
| | b) | With relevant sketches explain various Capacitors available in Analog CMOS Process. | 10 | L2 | 5 | 1 |
| 8. | a) | With neat diagrams explain the working of True Single-Phase Clocking (TSPC) based delay element. | 10 | L2 | 5 | 1 |
| | b) | Draw the circuit diagram for a 1-bit Full adder implemented using dynamic logic and explain its working. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
